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Case 27-2020: A 53-Year-Old Woman with Headache and Gait Imbalance

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PRESENTATION OF CASE

Dr. Jacob M. Rosenberg: A 53-year-old woman was transferred to this hospital because of progressive headache, gait instability, and weight loss.

Approximately 2 months before the patient's evaluation at this hospital, vertigo developed and resolved spontaneously. One day before evaluation at this hospital, she sought care at another hospital for recurrent dizziness and difficulty walking. The temperature was 36.4°C, the pulse 86 beats per minute, the blood pressure 170/105 mm Hg, the respiratory rate 15 breaths per minute, and the oxygen saturation 97% while she was breathing ambient air. The patient was alert and oriented, with slow speech, bilateral end-point horizontal nystagmus, and mild dysmetria on the right side on finger–nose–finger testing. Strength and sensory function were normal. She had difficulty maintaining a stable stance; her gait was narrow-based and described as “slow and cautious.” The remainder of the examination was normal.

Computed tomography (CT) of the head, performed without the administration of intravenous contrast material, showed an area of ill-defined hypoattenuation that was consistent with vasogenic edema in the right cerebellar hemisphere, as well as regions of white-matter hypoattenuation in both cerebellar hemispheres. Aspirin and atorvastatin were administered.

Dr. R. Gilberto Gonzalez: Magnetic resonance imaging (MRI) of the head (Fig. 1), performed before and after the administration of intravenous contrast material, revealed an enhancing intraaxial lesion, measuring 2.0 cm by 1.8 cm by 1.7 cm, in the right inferior cerebellar hemisphere. The lesion had internal signal hypointensity that was indicative of necrosis on T1-weighted imaging, as well as surrounding signal hyperintensity that was suggestive of vasogenic edema on T2-weighted imaging. There was local mass effect with effacement of the overlying cerebellar fissures and the fourth ventricle. There was also an enhancing lepto-

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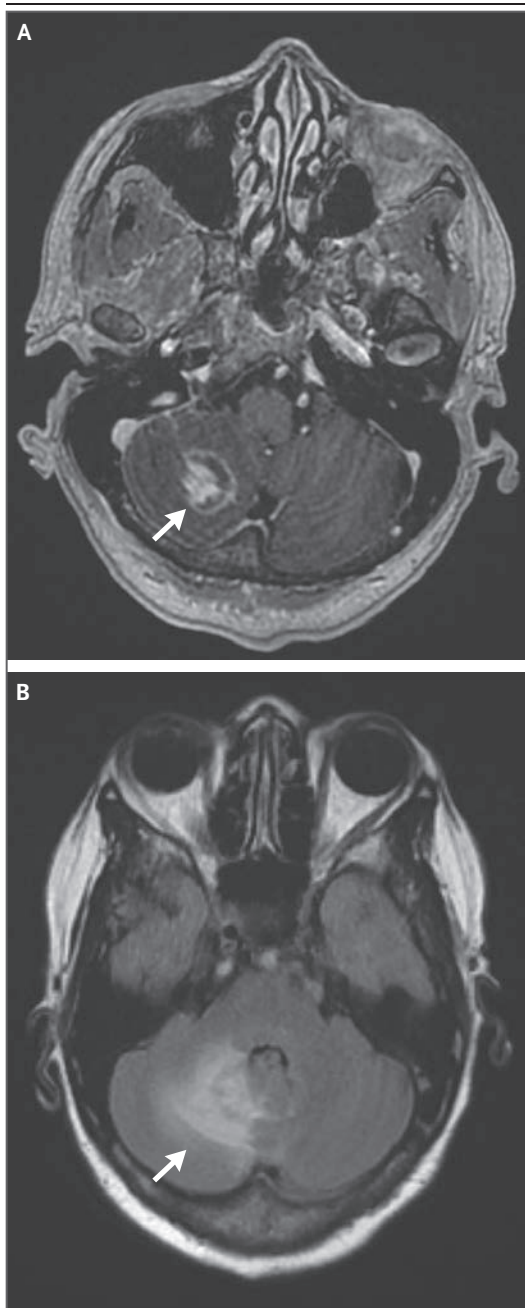


Figure 1. MRI of the Head.

An axial, T1-weighted image obtained after the administration of contrast material (Panel A) and a T2-weighted, fluid-attenuated inversion recovery image (Panel B) at the level of the fourth ventricle show a ring-enhancing abnormality with a nodular component (Panel A, arrow) and surrounding signal hyperintensity suggestive of vasogenic edema (Panel B, arrow) in the right inferior cerebellar hemisphere.

meningeal lesion, measuring 4 mm in diameter, along the left medial frontal lobe in the left superior frontal gyrus, with possible leptomeningeal extension. The lesion was characterized by internal and surrounding signal hyperintensity on T2-weighted imaging.

Dr. Rosenberg: Intravenous dexamethasone, amlodipine, and labetalol were administered. The patient was transferred to this hospital for specialized neurosurgical consultation.

On evaluation at this hospital, review of systems was notable for malaise and lethargy, as well as unintentional weight loss of 15 kg during the past year. The patient also reported an increase in urinary frequency without hematuria, dysuria, or flank pain. She had no fever, chills, sinus pain, dyspnea, cough, abdominal pain, diarrhea, nausea, or vomiting. She had a history of hypertension and hemorrhagic stroke, which had led to a craniotomy on the right side for evacuation of a “blood clot” 13 years before the current evaluation, with no reported neurologic deficits.

The patient reported that, during the admission for craniotomy, she was told that she had human immunodeficiency virus (HIV) infection and was treated with an unknown antiretroviral therapy (ART) regimen; she could not recall the medications included in the regimen. In addition, she reported that, several years later, further testing revealed that an error had been made and she did not have HIV infection. She reported that ART medications were discontinued at least 3 years before the current evaluation.

The patient did not have a history of opportunistic infections or cancer. Three years earlier, a Mantoux tuberculin skin test had been negative. One week earlier, a mammogram had been normal. She had never undergone a colonoscopy or a Papanicolaou test.

The patient was born in Ivory Coast, West Africa, and had been living in the United States for more than 20 years. She reported no recent travel. She lived with her children and did not smoke tobacco, drink alcohol, or use illicit drugs. She was divorced and not sexually active; in her lifetime, she had had one male sexual partner. Her ex-husband and one of her children reportedly had tested negative for HIV. Her father had had a myocardial infarction in his 60s;

her mother and sisters were healthy. The patient had not been taking any medications at home before this illness and had no known allergies to medications.

On physical examination, the temperature was 36.9°C, the blood pressure 165/99 mm Hg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 22.5. The patient was comfortable, alert, and oriented. Her speech was clear, with intact naming and repetition. Her pupils were equal and reactive to light, extraocular movements were intact, and visual fields were full in response to confrontation. She had mild right esotropia and no nystagmus. Her face was symmetric, but there was mild deviation of the tongue to the left. Results of other cranial-nerve testing were normal. Finger–nose–finger testing revealed subtle dysmetria that was greater on the right side than on the left side; there was no dysidiadochokinesia or intention tremor. Her stance was approximately shoulder-width. Her balance was impaired, and her gait was unsteady and ataxic. Psychomotor slowing was present, and the Romberg sign was absent. An evaluation of power, bulk, tone, sensation, and deep-tendon reflexes was normal. She had alopecia and oral thrush. The remainder of the examination was normal. The patient was admitted to the hospital for further evaluation and treatment.

Results of tests of renal, liver, and thyroid function were normal, as were blood levels of electrolytes, vitamin B₁₂, and glycated hemoglobin. An HIV screening test for the detection of antibodies to HIV type 1 (HIV-1) and HIV type 2 (HIV-2) and HIV-1 p24 antigen was reactive. Plasma HIV-1 RNA was not detected. The CD4+ T-cell count was 39 per microliter (reference range, 348 to 1456). Other laboratory test results are shown in Table 1. A urine sample was obtained for culture. CT of the chest, performed after the administration of intravenous contrast material, revealed a mildly enlarged and multinodular left thyroid gland, as well as scattered pulmonary nodules measuring less than 4 mm in diameter. CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed a hypoattenuating hepatic lesion measuring 1.1 cm in the greatest dimension that was consistent with hemangio-

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	On Admission
Blood		
Hemoglobin (g/dl)	12.0–16.0	13.9
Hematocrit (%)	36.0–46.0	42.9
Platelet count (per μ l)	150,000–400,000	232,000
White-cell count (per μ l)	4500–11,000	12,460
Differential count (per μ l)		
Neutrophils	1800–7700	11,101
Lymphocytes	1000–4800	685
Monocytes	200–1200	560
Eosinophils	0–900	0
Metamyelocytes	0	112
Urine		
Color	Yellow	Yellow
Clarity	Clear	Cloudy
pH	5.0–9.0	7.0
Specific gravity	1.001–1.035	1.021
Protein	Negative	1+
Glucose	Negative	Negative
Ketones	Negative	Trace
Bilirubin	Negative	Negative
Nitrite	Negative	Negative
Leukocyte esterase	Negative	3+
Red cells (per high-power field)	0–2	0–2
Leukocytes (per high-power field)	<10	50–100
Bacteria	None	2+

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

Table 2. Differential Diagnosis of the HIV Test Results.

HIV-negative with a false positive serologic screening test
HIV-1–positive with “elite controller” status and slow clinical progression
HIV-1–positive with a rare HIV-1 group (N, O, or P), HIV-1 group M subtype, or circulating recombinant form that is not detected by current HIV-1 viral load tests
HIV-negative with idiopathic CD4+ T-cell lymphopenia
HIV-1–positive with surreptitious use of antiretroviral therapy
Preanalytical error (sample mix-up)
HIV-2–positive

ma, as well as diffuse lytic osseous lesions in the thoracolumbar spine and pelvis.

Diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Geoffrey S. Gottlieb: This 53-year-old woman, originally from West Africa, presented with subacute, progressive focal neurologic symptoms, cerebellar signs, and an MRI of the head showing two contrast-enhancing lesions. She reported that, 13 years before the current admission, she received a diagnosis of HIV and was treated with an unknown ART regimen, and then several years later, testing revealed an error and ART was discontinued. On the current admission, a screening assay for antibodies to HIV-1 and HIV-2 and HIV-1 p24 antigen was reactive, and the CD4+ T-cell count was quite low at 39 per microliter; however, the plasma HIV-1 RNA level was undetectable. The first step in developing a differential diagnosis in this case is to interpret the confusing HIV test results and to determine whether the patient has HIV infection (Table 2).¹

HIV TEST RESULTS

Several clinical scenarios could yield the pattern of HIV test results seen in this case. The most common scenario is that the HIV antigen–antibody screening test had a false positive result. Although false positive results were more common with earlier versions of HIV antibody tests, current HIV antigen–antibody screening tests have a high specificity, with a false positive rate of no more than 40 per 10,000 tests, which makes a false diagnosis of HIV infection possible but unlikely. Furthermore, the patient's low CD4+ T-cell count would need to be explained. She could have a false positive HIV screening test and idiopathic CD4+ T-cell lymphopenia, a rare entity with an unclear cause, but this would be an unusual and unlikely combination of clinical events.

It is possible that the patient has HIV-1 infection but is an “elite controller,” a person who has natural control of HIV infection without ART but with slow clinical progression. This possibility would explain the positive HIV screening test and the undetectable HIV-1 RNA level. However, elite controller status is rare, with a prevalence of approximately 1 in 500 to 1000

persons with HIV-1 infection. Nonetheless, slow, decades-long disease progression has been observed in some elite controllers. This patient is 53 years old, so if she had become infected with HIV-1 in the 1980s, the infection could have progressed to acquired immunodeficiency syndrome (AIDS) by now.

It is also possible that this patient is infected with a rare HIV-1 group (N, O, or P), HIV-1 group M subtype, or circulating recombinant form that is not detected by current HIV-1 viral load tests. These strains are exceedingly rare but are known to be circulating in West and Central Africa, and this patient was originally from Ivory Coast.

Other potential explanations include surreptitious use of ART due to either stigma or neuropsychiatric issues. A preanalytical error, such as a mix-up of the blood sample for confirmatory testing or HIV-1 viral load testing, could have occurred. Finally, the patient could have infection with HIV-2, which is prevalent in West Africa and accounted for the majority of HIV cases in the 1980s and 1990s.

The diagnosis of HIV that the patient had received 13 years before this admission might have been correct. The details of previous testing and ART are not available currently but should be investigated. On the basis of the features of this patient's presentation, her positive HIV screening test in the context of an undetectable HIV-1 RNA level, and her country of origin, I suspect she has HIV-2 infection. Given the low CD4+ T-cell count, the presentation is consistent with HIV-2–associated AIDS.

ESTABLISHING THE DIAGNOSIS OF HIV-2 INFECTION

In this patient, I would recommend performing supplemental testing with an HIV-1 and HIV-2 antibody differentiation assay in accordance with Centers for Disease Control and Prevention guidelines,¹ as well as an HIV-2 viral load test to obtain the plasma HIV-2 RNA level. If these tests return negative, I would consider sending a whole-blood sample for nucleic acid testing for HIV-2 DNA and RNA. In the areas of West Africa in which HIV-2 is endemic, approximately one third of HIV-2–infected persons are coinfecting with HIV-1²; in this patient, the undetectable plasma HIV-1 RNA level essentially rules out HIV-1 coinfection.

NEUROLOGIC SYMPTOMS AND IMAGING FINDINGS

The next step in this case is to consider the patient's neurologic symptoms and imaging findings in the context of a likely diagnosis of HIV-2–associated AIDS. The differential diagnosis for brain lesions in people with HIV and AIDS is quite broad.³ Brain lesions may be mass or non-mass lesions, and they may or may not have contrast enhancement. Mass lesions of the brain may be caused by *Toxoplasma gondii* encephalitis (cerebral toxoplasmosis); tuberculosis with central nervous system (CNS) involvement (tuberculoma); primary CNS lymphoma, which is associated with Epstein–Barr virus infection; *Cryptococcus neoformans* infection with CNS involvement (cryptococcoma); brain abscess due to bacterial or fungal infection; neurocysticercosis; or *Treponema pallidum* infection (syphilitic gumma). Progressive multifocal leukoencephalopathy, due to JC virus and HIV encephalopathy, does not typically cause mass lesions. Cytomegalovirus encephalitis typically occurs without mass lesions and rarely causes mass effect. Of course, metastatic cancer to the brain and conditions unrelated to HIV and AIDS that affect the brain can lead to neurologic symptoms and imaging findings similar to those seen in this case.

Among all these possibilities, my leading diagnoses in this case are *T. gondii* encephalitis, tuberculoma, and primary CNS lymphoma. Several epidemiologic, clinical, laboratory, and imaging features of this case favor the diagnosis of *T. gondii* encephalitis. First, toxoplasmosis is the most common CNS infection among patients with AIDS who are not receiving prophylaxis, and this patient was not receiving trimethoprim–sulfamethoxazole. Historically, the 12-month incidence of *T. gondii* encephalitis among patients with AIDS who are not receiving prophylaxis has been approximately 33%. Second, this patient could have had exposure to toxoplasma in Ivory Coast, where the seroprevalence is 60 to 70%. Third, her CD4+ T-cell count is less than 100 per microliter, the level below which *T. gondii* encephalitis typically occurs, and this finding places her at high risk for the infection. Finally, her MRI lesions have contrast enhancement, as well as signal hypointensity indicative of necrosis on T1-weighted imaging. She has at least two CNS lesions with mass effect, and one is ring-enhancing. These features are all consistent with *T. gondii* encephalitis.

However, several features of this patient's case do not fit with the diagnosis of *T. gondii* encephalitis. She has had relatively slow clinical progression and has not reported a fever. *T. gondii* encephalitis with cerebellar involvement is not common. She has scattered pulmonary nodules (measuring <4 mm) of an unknown cause; this finding would not typically be seen in reactivation of *T. gondii* encephalitis but could be due to an unrelated condition. Her history of a cerebrovascular accident 13 years earlier could be related to her current illness, but given the time frame, I think a connection is unlikely.

ESTABLISHING THE DIAGNOSIS OF *T. GONDII* ENCEPHALITIS

To determine whether this patient has *T. gondii* encephalitis, I would perform serologic testing for IgM and IgG against toxoplasma. I would also start empirical treatment, perform ongoing clinical assessments, and perform repeat MRI of the head in 10 to 14 days. If she does not have evidence of improvement at that time, then I would pursue a brain biopsy, which is the standard approach for the diagnosis of brain lesions. Because mass effect is present on imaging in this patient, a lumbar puncture is probably not safe. However, if it were safe to obtain cerebrospinal fluid, then I would obtain a specimen for *T. gondii* nucleic acid testing, which has a specificity of 96 to 100% and a sensitivity of 50%.

**DR. GEOFFREY S. GOTTLIEB'S
DIAGNOSIS**

Toxoplasma gondii encephalitis in the context of human immunodeficiency virus type 2 infection and associated acquired immunodeficiency syndrome.

DIAGNOSTIC TESTING

Dr. Rosenberg: The patient had a reactive HIV antigen–antibody screening test, and a specimen was sent to an outside reference laboratory for an HIV-1 and HIV-2 antibody differentiation assay. HIV-2–specific bands at glycoproteins 36 and 140 were present on the HIV-2 differentiation assay, a finding that confirmed the diagnosis of HIV-2 infection. Nucleic acid testing subsequently showed a plasma HIV-2 RNA level of 486 cop-

Table 3. Management of HIV-2 Infection.*

HIV-2 infection is generally characterized by a longer asymptomatic stage, lower plasma viral loads, and a lower mortality than HIV-1 infection. However, progression to AIDS and death occur in the majority of patients who do not receive treatment.
No randomized, controlled trials have addressed when to start ART or the choice of initial or second-line ART for HIV-2 infection. Thus, the appropriate treatment strategy is not well defined.
Data on the treatment of HIV-2 and extrapolation from data on the treatment of HIV-1 suggest that ART should be started at or soon after the diagnosis of HIV-2 to prevent disease progression and transmission of HIV-2 to others.
Quantitative HIV-2 viral load testing for clinical care is available and should be performed before the initiation of ART.
HIV-2 is intrinsically resistant to NNRTIs and to enfuvirtide. Thus, these drugs should not be included in ART regimens for HIV-2 infection.
Hepatitis B virus infection should be considered when choosing ART for HIV-2 infection, as for HIV-1 infection.
Initial ART regimens for patients who have not previously received ART and have HIV-2 monoinfection or HIV-1 and HIV-2 coinfection should include an INSTI plus two NRTIs. An alternative regimen for these patients is an HIV-2–active, boosted PI plus two NRTIs.
HIV-2 RNA, CD4+ T-cell counts, and clinical status should be used to assess treatment response. Unlike persons with HIV-1, persons with HIV-2 should continue to undergo periodic monitoring of the CD4+ T-cell count, even if the viral load is persistently suppressed, since disease progression can occur despite an undetectable viral load.
Viral mutations that confer resistance to INSTIs, PIs, or NRTIs may occur in persons with HIV-2 while they are receiving ART. However, no validated HIV-2 genotypic or phenotypic drug-resistance assays are approved for clinical use.
In the event of virologic, immunologic, or clinical failure, a new ART regimen should be constructed in consultation with an expert in HIV-2 management.

* Data are adapted from the Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents.³ ART denotes antiretroviral therapy, INSTI integrase strand-transfer inhibitor, NRTI nucleoside reverse-transcriptase inhibitor, NNRTI nonnucleoside reverse-transcriptase inhibitor, and PI protease inhibitor.

ies per milliliter. The day after the patient was admitted to this hospital, *T. gondii* serologic testing was reported as positive for IgG and negative for IgM, confirming a history of toxoplasmosis and supporting the clinical diagnosis of *T. gondii* encephalitis.

MANAGEMENT OF HIV-2 INFECTION

Dr. Gottlieb: Guidelines for the management of HIV-2 infection have recently been updated by the Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents (Table 3).³ How should we man-

age HIV-2 infection in this patient? Strategies that are recommended for the management of HIV-1 infection — including HIV primary care, prophylaxis for opportunistic infections, screening for infections, and vaccinations — should be pursued.⁴

Approximately 8 to 10% of HIV-infected West Africans have chronic hepatitis B virus (HBV) infection, and some of these patients are co-infected with hepatitis D virus (HDV), so reflexive screening for HDV is warranted if HBV is detected. This patient had not been to Ivory Coast for more than 20 years and thus is unlikely to have a chronic parasitic disease, although there is a possibility of strongyloidiasis, for which I would screen and consider empirical ivermectin treatment. I would also screen for human T-cell lymphotropic virus infection, since coinfection with this virus and HIV is not uncommon in West Africa. Screening for latent tuberculosis was negative in this patient 3 years earlier, but given her low CD4+ T-cell count, she has probably become anergic. Once her acute illness is over, I would consider rescreening for tuberculosis with an interferon- γ release assay; the chest CT did not show any evidence of previous exposure to tuberculosis.

After treatment for *T. gondii* encephalitis, the patient would need to receive secondary prophylaxis for toxoplasmosis and primary prophylaxis for pneumocystis pneumonia until her CD4+ T-cell count is higher than 200 per microliter. According to updated guidelines, she would not need to receive prophylaxis for mycobacterial infection. It should be noted that, among patients with HIV-2 infection, immune reconstitution associated with ART is slower than it is among patients with HIV-1 infection⁵ and immune reconstitution inflammatory syndrome has been observed only rarely.

Given this patient's previous exposure to ART, I would perform an HIV-2 genotypic drug-resistance test to determine the presence of mutations that confer resistance to protease inhibitors (PIs), nucleoside reverse-transcriptase inhibitors (NRTIs), or integrase strand-transfer inhibitors (INSTIs), since we do not know which drugs she had received. In the United States, no Food and Drug Administration (FDA)–approved HIV-2 genotypic drug-resistance assays are available, but testing may be available in research

settings. It would also be appropriate to perform an HIV-2 genotypic drug-resistance test if there is virologic failure or failure to suppress the viral load by 6 months after initiation of ART.

A baseline plasma HIV-2 RNA level has already been obtained. I would recheck the level 4 to 6 weeks after starting ART, then every 3 months for the first year. Once the viral load has been undetectable for a year, I would perform viral load testing every 6 months. In addition, the patient's CD4+ T-cell count was very low (39 per microliter) at presentation. I would recheck the count 4 to 6 weeks after starting ART, then every 3 months until it is higher than 200 per microliter; thereafter, I would check it every 6 to 12 months. HIV-2 disease can progress without apparent virologic failure while a patient is receiving ART, so monitoring the CD4+ T-cell count periodically, even in patients with seemingly well-controlled disease, is a reasonable strategy.

It should be noted that no ART drugs have been approved by the FDA for the treatment of HIV-2 infection. The off-label use of the drugs is based on their approved indication for the treatment of HIV-1 infection. There have been no randomized, controlled trials of ART for HIV-2 infection, although one such trial is under way (ClinicalTrials.gov number, NCT02150993). On the basis of currently available, albeit limited, data (Table 3),³ I would initiate ART with an INSTI and two NRTIs, which would ideally be administered in a single tablet for ease of use and adherence.^{6,7} A fixed-dose combination of either abacavir–dolutegravir–lamivudine (if the patient is negative for HLA B5701 and HBV) or bictegravir–emtricitabine–tenofovir alafenamide would be a reasonable choice on the basis of in vitro and clinical data.^{8–13} Nonnucleoside reverse-transcriptase inhibitor–based regimens should be avoided because HIV-2 is intrinsically resistant to these drugs.¹⁴ A PI-based regimen could be considered as an alternative to an INSTI-based regimen. If a PI is used, then one with sufficient activity against HIV-2, such as boosted darunavir or lopinavir, should be chosen, since many of the HIV-1 PIs have reduced activity against HIV-2.^{4,15}

Finally, I would recommend that the patient's ex-husband, any sexual partners, and her children be tested for HIV-2.

MANAGEMENT OF BRAIN LESIONS

Dr. Rajesh T. Gandhi: A patient who has cerebral toxoplasmosis in addition to HIV and AIDS should be treated with antiparasitic therapy along with the prompt initiation of ART. Initial antiparasitic therapy consists of sulfadiazine and pyrimethamine; leucovorin is also administered to counteract the myelosuppressive effects of the treatment.¹⁶ In the United States, pyrimethamine has unfortunately become prohibitively expensive in many instances because of a large price increase by the company that acquired the marketing rights to the drug.¹⁷ If pyrimethamine cannot be obtained because of cost or lack of availability, then trimethoprim–sulfamethoxazole should be used instead of sulfadiazine and pyrimethamine.¹⁶

The appropriate timing for the initiation of ART in people with HIV and an acute opportunistic infection was evaluated in a randomized clinical trial.¹⁸ Among 282 people with a variety of opportunistic infections, early ART (given within 14 days after therapy for the opportunistic infection was started) was associated with a lower rate of AIDS progression or death than deferred ART (given after therapy for the opportunistic infection was completed). Although the trial included only 13 people with toxoplasmosis, it is still reasonable to extend the findings to a patient such as this one. In this patient, I would start HIV-2–active ART as soon as it is clear that she is receiving the antiparasitic treatment without complication, which would typically be within a few days to a week.

FOLLOW-UP

Dr. Rosenberg: On discharge from the hospital, the patient had moderate improvement of her subjective weakness, confusion, and gait instability. Her headache had resolved with management of her hypertension, and she had not had a fever. Medications on discharge included tenofovir alafenamide, emtricitabine, and dolutegravir for HIV-2 infection and pyrimethamine, sulfadiazine, and leucovorin for cerebral toxoplasmosis. However, at a 1-week follow-up appointment in the infectious diseases clinic, she was found to have a blood creatinine level of 3.5 mg per deciliter (309 μ mol per liter; reference range, 0.6 to 1.5 mg per deciliter [53 to 133 μ mol per liter]) and a

urine eosinophil count of 70%. She received a diagnosis of sulfadiazine-induced acute interstitial nephritis; sulfadiazine was discontinued, and a regimen of pyrimethamine, clindamycin, and leucovorin was started. Atovaquone was added for pneumocystis pneumonia prophylaxis. At a 3-month follow-up appointment, her renal function had improved and the HIV-2 viral load was undetectable. At a 6-month follow-up appointment, MRI of the head showed a marked decrease in the diameter of the ring-enhancing lesion in the right cerebellar hemisphere, as well as resolution of the ring-enhancing lesion of the left medial frontal lobe. One year after the initiation of ART, the CD4⁺ T-cell count was 203 per microliter. A discussion with the patient's previous health care provider revealed that the patient had been counseled to stop ART approximately 8 years earlier. In the context of a positive HIV screening test and a negative HIV-1 viral load test, the provider incorrectly interpreted the initial test results as false positive.

FINAL DIAGNOSIS

Human immunodeficiency virus type 2 infection and cerebral toxoplasmosis.

This case was presented at the 13th Annual Workshop on Advanced Clinical Care–AIDS in Durban, South Africa, organized by Drs. Henry Sunpath and Mahomed-Yunus S. Moosa (Infectious Diseases Unit, Nelson R. Mandela School of Medicine, University of KwaZulu-Natal) and Dr. Rajesh T. Gandhi (Massachusetts General Hospital and the Ragon Institute) and sponsored by the Harvard University Center for AIDS Research (NIH P30 AI060354), the Centre for the AIDS Programme of Research in South Africa, the University of KwaZulu-Natal, the Southern African HIV Clinicians Society, and the KwaZulu-Natal Department of Health.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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